Name

Mathematics Advanced Year 11

Sets and Set Notation

Book 1

Version: 261021

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Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

MAV-11-10 displays and analyses datasets using summary statistics and graphical representations

Sets and set notation

- Define a set as a collection of objects, called the elements of the set, and represent a set using notation such as $\{2,3,5,7\}$ and $\{0,2,4,6, \ldots\}$
- Use the notation $n(A)$ or $|A|$ to represent the number of elements in a finite set A
- Define the empty set as the set with no elements, denoted in set notation as $\emptyset$
- Use the notation $\bar{A}$, A' or A^c to represent the complement of a set A with respect to some universal set U
- Define A to be a subset of B if all the elements of A are elements of B
- Define the intersection $A \cap B$ of sets A and B to be the set of elements that are in A and in B
- Define the union $A \cup B$ of sets A and B to be the set of elements that are in A or in B
- Define sets A and B to be disjoint if $A \cap B = \emptyset$, that is, they have no elements in common
- Use Venn diagrams in practical situations to represent and interpret sets that may intersect in various ways within a universal set
- Establish and use the rule $|A \cup B| = |A| + |B| - |A \cap B|$

Set Notation

ⓘ Sets

A **set** is a collection of **elements** (objects).

We use { } to represent a set. Each element is separated by commas.

Sets have **no repeated elements** and **no order**.

The **empty set** has no elements, denoted in set notation as: $\emptyset$

Interpret set notation.

"x is the set of numbers 7, 3, 2, 14":

$x = \{7, 3, 2, 14\}$

"y is the set of numbers 6, 2, 2, and 7":

$y = \{6, 2, 7\}$

Use set notation to list the elements of each of these sets.

a $A =$ {even numbers between 10 and 20 inclusive}

b $B =$ {days of the week}

c $C =$ {letters in the word "integer"}

d $D =$ {negative real square numbers}

ⓘ Element of Symbol

$a \in X$ means a is included in set X.

"a is an element of $X = \{a,\ b,\ c\}$"

$a \in X$

Interpret element of symbol.

✗ $4 \notin \{1,2,3\}$ ✓ $2 \in \{1,2,3\}$ ✓ $\text{dog} \in \{\text{cat}, \text{dog}\}$ ✓ $32.5 \in \mathbb{R}$ ✗ $\sqrt{-3} \notin \mathbb{R}$

Write the correct symbol, $\in$ or $\notin$.

a 4 ..... $\{1, 2, 3\}$

b 1 ..... {12, 13, 15}

c $\sqrt{5}$ ..... $\mathbb{R}$

d red ..... {blue, black}

ⓘ Number of Elements

$n(A)$ or $|A|$ represents the number of elements inside set A.

Interpret number of elements notation.

$A = \{2,5,6,11\}$ $n(A) = 4$

$X = \{\text{dog, cat, wednesday}\}$ $n(X) = 3$

$S = \{\text{even prime numbers}\}$ $n(A) = 1$

Determine the number of elements in these sets.

a $A = \{9,8,23,77,-2\}$

$n(A) =$

b $B =$ {months of the year}

$|B| =$

c $S =$ {letters in the word "integer"}

$|S| =$

d $C =$ {prime numbers between 1 and 10}

$n(C) =$

ⓘ Universal Sets and The Complement

U or ξ is the **universal set** that contains all elements we're considering.

$\bar{A}$, A' or A^c represents the **complement** of set A.

The **complement** contains all elements that are in the universal set U but are NOT in set A.

A and A' together make up the universal set.

Interpret universal sets and complement sets.

$U = \{\text{Integers between 0 and 10, inclusive}\}$
$= \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

$A = \{\text{Even integers}\}$
$= \{0, 2, 4, 6, 8, 10\}$

$A' = \{\text{Not even integers}\}$
$= \{1, 3, 5, 7, 9\}$

$\xi = \{\text{Integers between 0 and 10, inclusive}\}$
$= \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

$A = \{\text{Multiples of 3}\}$
$= \{0, 3, 6, 9\}$

$A^c = \{\text{Not multiples of 3}\}$
$= \{1, 2, 4, 5, 7, 8, 10\}$

List the elements for these sets, given the universal set.

a $U = \{\text{days of the week}\}$

$A = \{\text{weekdays}\}$

$A' =$

b $U = \{\text{square numbers up to 100, inclusive}\}$

$B = \{\text{multiples of 5}\}$

$B' =$

c $\xi = \{\text{Factors of 12}\}$

$A = \{\text{even factors}\}$

$A^c =$

d $\xi = \{\text{Multiples of 4 up to 28, inclusive}\}$

$A = \{\text{odd multiples}\}$

$\bar{A} =$

ⓘ Subsets

A is a subset of B if every element in A is also in B.

"A is a subset of B"

$$A \subseteq B$$

⚠ Every set is a subset of itself: A ⊆ A
⚠ Empty set is subset of every set: ∅ ⊆ A
⚠ A ⊂ B means A ⊆ B AND A ≠ B

ⓘ **Identify** subsets.

✗	✓	✓	✓	✗
A = {1, 3, 7} B = {1, 3, 5}	A = {1, 3} B = {1, 3, 5}	A = {R, B, G} B = {R}	X = {4, 2, 8, 6} Y = {2, 4, 6, 8}	X = {2, 4, 0} Y = {2, 4, 6, 8}
$A \not\subset B$	$A \subseteq B$	$B \subseteq A$	$X \subseteq Y$ $Y \subseteq X$	$X \not\subset Y$

Write the correct symbol ⊆ or ⊄ to relate the two sets.

a	A = {2, 4} B = {1, 2, 3, 4, 5}	**b**	A = {a, c} B = {a, b, d}	**c**	A = {real numbers ℝ} B = {integers ℤ}
d	X = {3, 4, 5} Y = {5, 3}	**e**	A = {x, y, z} B = {z, y, x}	**f**	H = {2, 3} K = {6}

ⓘ Intersection of Sets

The intersection of sets A and B are the set of elements that are **both** in A **and** in B.

"Intersection of A and B"

$$A \cap B$$

Interpret intersections of sets.

A = {1, 3, 5, 7}	C = {red, blue, green}	E = {2, 4, 6}
B = {3, 5, 9, 11}	D = {blue, green, yellow}	F = {1, 3, 5}
A ∩ B = {3, 5}	C ∩ D = {blue, green}	E ∩ F = ∅

Identify the intersection of these sets.

a A = {1, 2, 3, 5, 8} B = {2, 3, 4, 7}	**b** C = {d, b, c, a} D = {c, d, e, f}	**c** C = {red, green, blue} D = {green, orange, purple}

ⓘ Mutually Exclusive or Disjoint Sets

Two sets A and B are **mutually exclusive** (or **disjoint**) if they have no elements in common.

The intersection of A and B is an empty set.

$$A \cap B = \emptyset$$

ⓘ **Identify** disjoint/mutually exclusive sets.

Not mutually exclusive	Mutually exclusive	Mutually exclusive	Mutually exclusive	Not mutually exclusive
A = {1, 3, 7} B = {1, 5}	A = {1, 3, 7} B = {2, 5}	A = {R, B, G} B = {Y}	X = {positive} Y = {negative}	X = {5, 6, 7} Y = {1, 7, 12}

Identify which of these pairs of sets are mutually exclusive.

a	A = {x, y} B = {a, b}	**b**	A = {a, c} B = {a, b, d}	**c**	A = {real numbers $\mathbb{R}$} B = {integers $\mathbb{Z}$}
d	X = {odd numbers} Y = {even numbers}	**e**	A = {prime numbers} B = {multiples of 3}	**f**	H = {humans} K = {mammals}

ⓘ Union of Sets

The union of sets A and B are the set of elements that are in **either** A **or** in B.

"Union of A and B"

$$A \cup B$$

Interpret unions of sets.

A = {1, 3}	C = {red, blue}	E = {a, b}
B = {2, 3, 6}	D = {blue, green}	F = {c, d}
A ∪ B = {1, 2, 3, 6}	C ∪ D = {red, blue, green}	E ∪ F = {a, b, c, d}

Identify the union of these sets.

a A = {1, 2} B = {3, 4}	**b** C = {1, 2, 3} D = {3, 4, 5}	**c** C = {1, 2, 3} D = {1, 2, 3}
d A = {1, 2} B = {1, 2, 3, 4}	**e** C = {red, blue} D = {green, blue, yellow}	**f** C = {1, 2, 3, 4} D = {4, 1, 7, 5}

Example interpret set notation.

Universal set $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
Set $A = \{2, 4, 6, 8\}$
Set $B = \{1, 2, 3, 4\}$
Set $C = \{5, 7, 9\}$

For each question, write the meaning in English and give the answer.

a A ∪ B Elements in either A or B = {1, 2, 3, 4, 6, 8}	**b** A ∩ B Elements in both A and B = {2, 4}
c B ∪ C Elements in either B or C = {1, 2, 3, 4, 5, 7, 9}	**d** A ∩ C Elements in both A and C = ∅
e n(A) Number of elements in A = 4	**f** |C| Number of elements in C = 3
g n(A ∪ B) Number of elements in A or B = 6	**h** A' Complement of A = {1, 3, 5, 7, 9, 10}
i $\overline{B}$ Complement of B = {5, 6, 7, 8, 9, 10}	**j** Which sets are mutually exclusive? A and C

True or false?

k A ⊆ B? A subset of B? False. (6 and 8 are in A but not in B)	**l** B ⊆ A? B subset of A? False. (1 and 3 are in B but not in A)
m {2, 4} ⊆ A? {2, 4} subset of A? True. (both 2 and 4 are elements of A)	**n** B ⊆ U? B subset of the universal set? True.

Find the following by writing out each set combination separately.

o (A ∪ B) ∩ C {1, 2, 3, 4, 6, 8} ∩ {5, 7, 9} = ∅	**p** A ∩ B' {2, 4, 6, 8} ∩ {5, 6, 7, 8, 9, 10} = {6, 8}
q (A ∪ C)' {2, 4, 5, 6, 7, 8, 9}' = {1, 3, 10}	**r** n(A ∪ C)' 3

1 Geri asks 10 students in her class to name their favourite colour. Geri writes the results as set A.
A = {green, purple, pink, red, blue, red, red, green, orange, red}.
Explain the mistake that Geri has made.

Sets cannot contain duplicate elements - "red" appears 4 times

2 Write down the following sets by listing their members.

a Odd positive integers less than 10.

{1, 3, 5, 7, 9}

b The first six positive multiples of 6.

{6, 12, 18, 24, 30, 36}

c The numbers on a die.

{1, 2, 3, 4, 5, 6}

d The factors of 20.

{1, 2, 4, 5, 10, 20}

3 List the elements of each of these sets. Use correct set notation.

a X = {multiples of 7 between 20 and 40}

X = {21, 28, 35}

b Y = {seasons in a year}

Y = {spring, summer, autumn, winter}

c Z = {letters in the word "science"}

Z = {s, c, i, e, n}

4 ξ = {2, 4, 6, 8, 10, 12, 14},
A = {2, 4, 8, 12}.
Write down the elements of A'.

A' = {6, 10, 14}

5 A = {10, 11, 12, 13, 14, 15},
B = {12, 13, 14}.
List the elements of A ∩ B.

A ∩ B = {12, 13, 14}

6 A = {red, white, blue},
B = {green, red, yellow, black}.
List the elements of A ∩ B.

A ∩ B = {red}

7 If A = {1, 3, 5}, B = {3, 4} and E = {1, 2, 3, 4, 5}, find:

a |A|

b |B|

|A| = 3

|B| = 2

8 ξ = {numbers from 1 to 50},
A = {odd numbers},
B = {even numbers},
C = {multiples of 10},
D = {square numbers}.

a List all the elements of set D.

D = {1, 4, 9, 16, 25, 36, 49}

b List all the elements that are in both set B and set D.

B ∩ D = {4, 16, 36}

c Explain why there are no elements that are listed in both set A and set C.

All multiples of 10 are even, but A contains only odd numbers

d List the elements that are in set C or D or both.

C ∪ D = {1, 4, 9, 10, 16, 20, 25, 30, 36, 40, 49}

9 A = {factors of 12}, B = {factors of 30}.

a List the elements of set A.

A = {1, 2, 3, 4, 6, 12}

b List the elements of set B.

B = {1, 2, 3, 5, 6, 10, 15, 30}

c Which numbers are in both set A and set B?

A ∩ B = {1, 2, 3, 6}

10 Mateusz and Hannah are writing down as many three-letter words beginning with 't' as possible in 30 seconds.
M = {tab, tan, tam, the, tea, ten},
H = {tap, tax, tan, the, ten}.

a List the elements of M ∪ H.

M ∪ H = {tab, tan, tam, the, tea, ten, tap, tax}

b Describe what M ∪ H tells you.

b) All three-letter words beginning with 't' that either student wrote

11 ξ = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10},
A = {2, 4, 6, 8, 10},
B = {5, 10},
C = {2, 3, 5, 7}. Write down the elements of each of these sets.

a A ∪ B

A ∪ B = {2, 4, 5, 6, 8, 10}

b A ∪ C

A ∪ C = {2, 3, 4, 5, 6, 7, 8, 10}

c B ∪ C

B ∪ C = {2, 3, 5, 7, 10}

12 X = {multiples of 5 between 12 and 33},
Y = {even numbers}.
List the elements of X ∩ Y.

X = {15, 20, 25, 30}, Y = {even numbers}

X ∩ Y = {20, 30}

13 ξ = {multiples of 5 between 1 and 49},
A = {multiples of 3},
B = {multiples of 2}.

a Write down the elements of A ∪ B.

b Write down the elements of A ∩ B.

ξ = {5, 10, 15, 20, 25, 30, 35, 40, 45}

A ∪ B = {10, 15, 20, 30, 40, 45}

A ∩ B = {30}

14 ξ = {e, g, h, m, n, p, s, t},
A = {h, p, s, t},
B = {e, g, h, m, p, t}.
Write down the elements of each of these sets.

a A'

A' = {e, g, m, n}

b B'

B' = {n, s}

c (A ∩ B)'

A ∩ B = {h, p, t}, so (A ∩ B)' = {e, g, m, n, s}

d (A ∪ B)'

A ∪ B = {e, g, h, m, p, s, t}, so (A ∪ B)' = {n}

15 ξ = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10},
A = {2, 4, 6, 8, 10},
B = {5, 10},
C = {2, 3, 5, 7}.
Write down the elements of each of these sets.

a B ∪ C

B ∪ C = {2, 3, 5, 7, 10}

b (A ∪ B) ∪ C

(A ∪ B) ∪ C = {2, 3, 4, 5, 6, 7, 8, 10}

c Is (A ∪ B) ∪ C the same as (A ∪ C) ∪ B?

Yes

16 If A = {1, 3, 5}, B = {3, 4} and universal set U = {1, 2, 3, 4, 5}, find:

a $A \cup B$

$A \cup B = \{1, 3, 4, 5\}$

b $|A \cup B|$

$|A \cup B| = 4$

c $A \cap B$

$A \cap B = \{3\}$

d $|A \cap B|$

$|A \cap B| = 1$

e $\bar{A}$

$\bar{A} = \{2, 4\}$ (complement in U)

f $\bar{B}$

$\bar{B} = \{1, 2, 5\}$ (complement in U)

17 If A = {1, 4, 7, 8} and B = {1, 2, 4, 5, 7}, state whether the following statements are true or false.

a $A = B$

False

b $|A| = 4$

True

c $|B| = 6$

False

d $A \subset B$

False, 8 is in A but not B

e $A \cup B = \{1, 2, 4, 5, 7, 8\}$

True

f $A \cap B = \{1, 4, 7\}$

True

18 A = {red, white, blue}, B = {green, red, yellow, black}.
Explain why $A \cup B$ is not {red, white, blue, green, red, yellow, black}.

sets cannot contain duplicates - "red" would only appear once

19 State in each case whether or not $A \subset B$ (that is, whether A is a subset of B).

a A = {2, 3, 5}, B = {1, 2, 3, 4, 5, 6, 7, 8, 9}

True

b A = {3, 6, 9, 12}, B = {3, 5, 9, 11}

False (6, 12 not in B)

c A = {d, a, n, c, e}, B = {e, d, u, c, a, t, i, o, n}

True

d A = {a, m, y}, B = {s, a, r, a, h}

False (m, y not in B)

e $A = \emptyset$, B = {51, 52, 53, 54}

True (empty set is subset of any set)

20 Decide which elements would need to be removed from event A if the two events A and B described below are to become mutually exclusive.

a A = {1, 2, 3, 4}, B = {4, 5, 6, 7}

Remove 4 from A

b A = {10, 12, 14, 16, 18}, B = {9, 10, 11, 12}

Remove 10, 12 from A

c A = {a, b, c, d, e}, B = {a, c, e, g}

Remove a, c, e from A

d A = {1, 3, 5, 8, 10, 15, 20, 22, 23}, B = {7, 9, 14, 16, 19, 21, 26}

No elements need removing (already mutually exclusive)

21 If A = {students who study Japanese} and B = {students who study History}, carefully describe each of the following sets.

a $A \cap B$

Students who study both Japanese and History

b $A \cup B$

Students who study Japanese or History (or both)

22 Find $A \cup B$ and $A \cap B$ for each pair of sets.

a A = {1, 3, 5}, B = {3, 5, 7}

$A \cup B = \{1, 3, 5, 7\}$, $A \cap B = \{3, 5\}$

b A = {1, 3, 4, 8, 9}, B = {2, 4, 5, 6, 9, 10}

$A \cup B = \{1, 2, 3, 4, 5, 6, 8, 9, 10\}$, $A \cap B = \{4, 9\}$

c A = {h, o, b, a, r, t}, B = {b, i, c, h, e, n, o}

$A \cup B = \{h, o, b, a, r, t, i, c, e, n\}$, $A \cap B = \{h, o, b\}$

d A = {j, a, c, k}, B = {e, m, m, a}

$A \cup B = \{j, a, c, k, e, m\}$, $A \cap B = \{a\}$

e A = {prime numbers less than 10}, B = {odd numbers less than 10}

A = {2, 3, 5, 7}, B = {1, 3, 5, 7, 9}

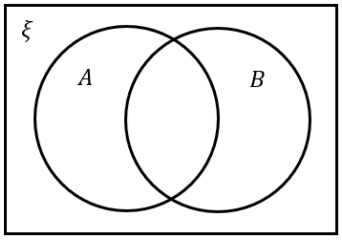
$A \cup B = \{1, 2, 3, 5, 7, 9\}$, $A \cap B = \{3, 5, 7\}$

Visualising Sets using Venn Diagrams

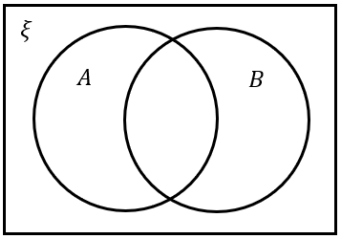
Interpret set notation using Venn diagrams.

Shade the region in the Venn diagram that represents the given set.

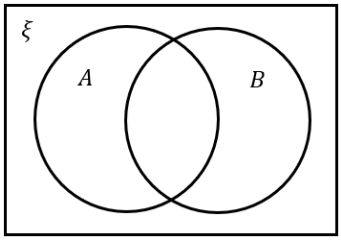
a A



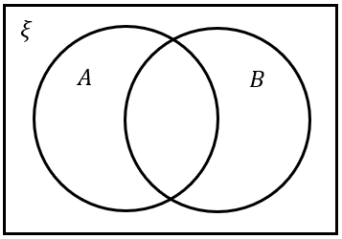
b B



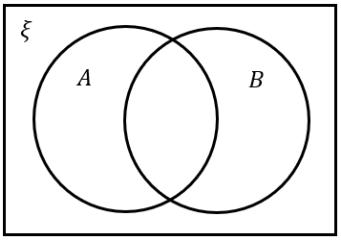
c $A \cap B$



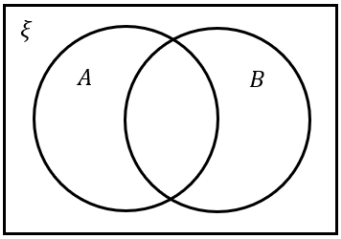
d $A \cup B$



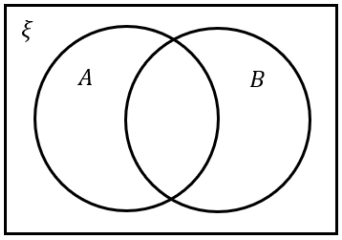
e $\bar{A}$



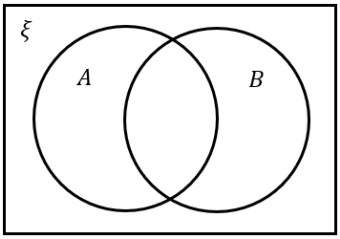
f $\bar{B}$



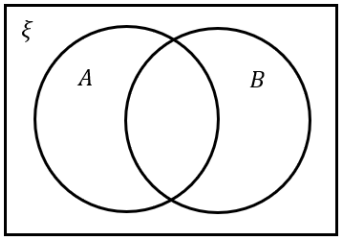
g $\bar{A} \cap \bar{B}$



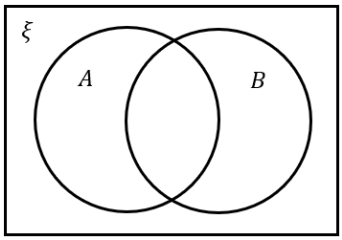
h $\bar{A} \cap B$



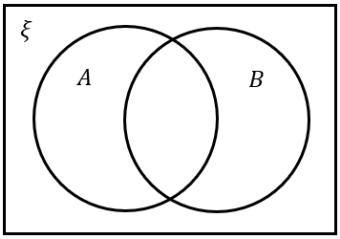
i $A \cap \bar{B}$



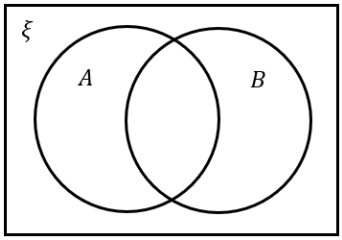
j $\overline{(A \cup B)}$



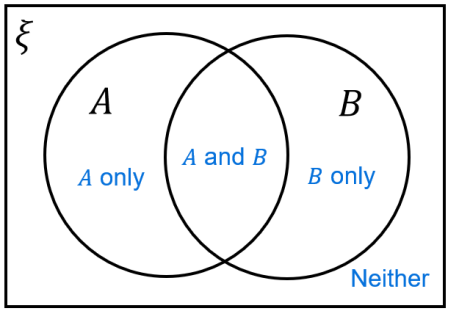
k ξ



l $(A \cup B) - (A \cap B)$



Venn Diagrams and Sets



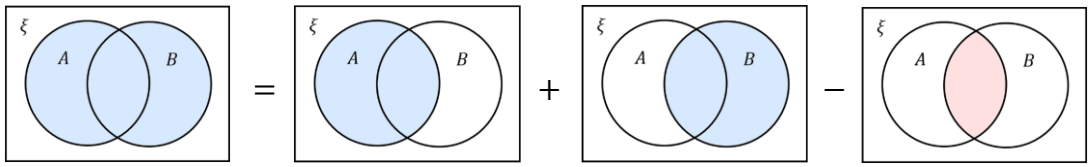
ⓘ Addition Rule for Sets

Imagine you're counting students who study maths or science.

If you add the students who study maths and students who study science, you will double-count students who study both.

So, you must subtract the people who study both.

$$|A \cup B| = |A| + |B| - |A \cap B|$$



Example Apply addition rule for sets.

In a sports club, 22 play basketball, 18 play volleyball, and 8 play both sports. How many members play basketball or volleyball?

$$|V \cup B| = |V| + |B| - |V \cap B|$$
$$= 22 + 18 - 8$$
$$= 32$$

A survey of 80 teenagers found that 35 like pop music, 28 like rock music, and 12 like both genres. How many teenagers like pop or rock music?

$$|P \cup R| = |P| + |R| - |P \cap R|$$
$$= 28 + 12 - 12$$
$$= 51$$

Practice

In a school of 120 students, 45 study Spanish, 52 study French, and 15 study both languages. How many students study Spanish or French?

82

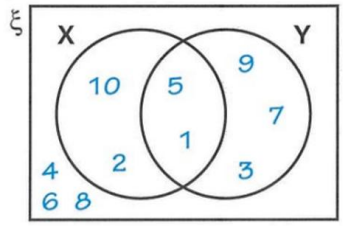
$A = \{\text{whole numbers from 1 to 10 inclusive}\}$

$B = \{\text{even numbers from 1 to 15}\}$

Find $|A \cup B|$ without listing the set.

12

1



a List the elements of set X.

$X = \{10, 5, 2, 1\}$

b List the elements of set Y.

$Y = \{9, 5, 1, 7, 3\}$

c List the elements of both set X and set Y.

$X \cap Y = \{5, 1\}$

d List the elements of set X or set Y.

$X \cup Y = \{10, 5, 2, 1, 9, 7, 3\}$

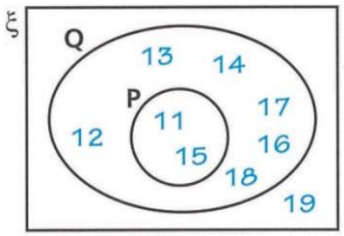
e List the elements that are not in set X.

$\bar{X} = \{4, 6, 8, 9, 7, 3\}$

f List the elements that are not in set Y.

$\bar{Y} = \{10, 2, 4, 6, 8\}$

2



a List the elements of set P.

$P = \{11, 5\}$

b Sven says that 11 is not an element of set Q. Is Sven correct?

Sven is incorrect. 11 is an element of Q

c Write down the elements of set Q.

$Q = \{12, 13, 14, 17, 16, 18, 11, 15\}$

d How many elements are not in set Q?

1

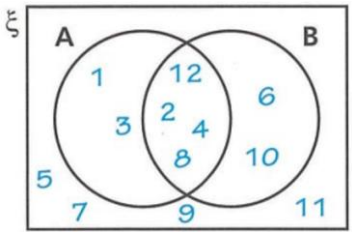
e How many elements are not in set P?

7

f Explain why the set “both P and Q” are the same as those of set P.

P ⊆ Q (P is subset of Q)

3



a List the elements of A ∩ B.

$A \cap B = \{12, 2, 4, 8\}$

b List the elements of A ∪ B.

$A \cup B = \{1, 3, 2, 12, 4, 8, 6, 10\}$

c List the elements of A'.

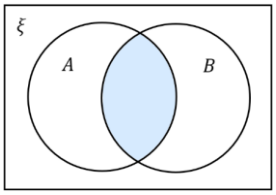
$A' = \{5, 7, 9, 6, 10, 11\}$

d List the elements of B'.

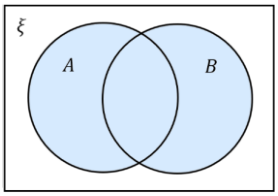
$B' = \{1, 3, 5, 7, 9, 11\}$

4 Shade the region:

A ∩ B

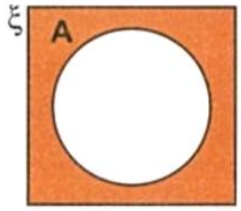


A ∪ B

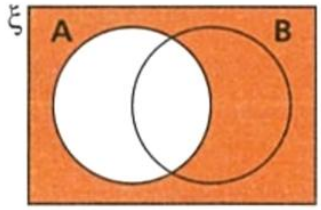


5 On each diagram, shade the region A'.

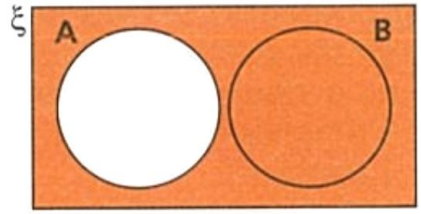
a



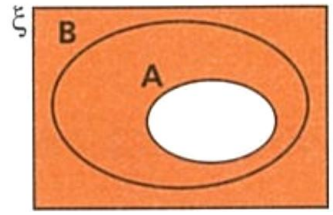
b



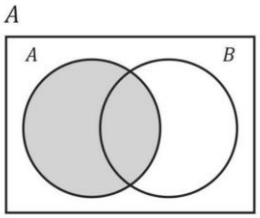
c

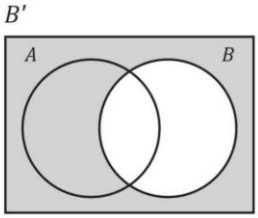


d

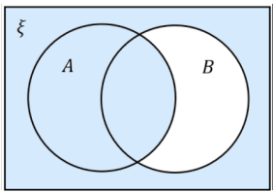


6 The two Venn diagrams show A and B'.

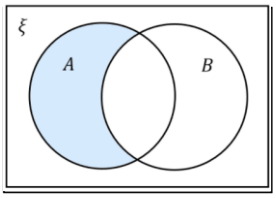




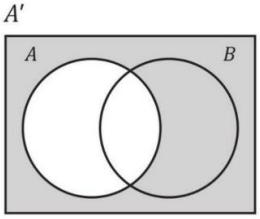
a Shade $A \cup B'$ by shading everything that is either in the first picture or in the second picture.

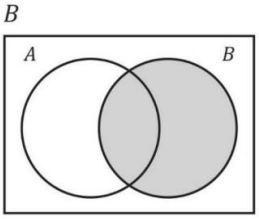


b Shade $A \cap B'$ by shading anything that is in both the first and second picture

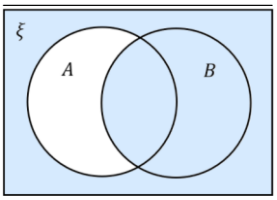


7 The two Venn diagrams show A' and B.

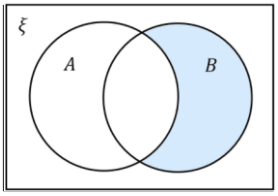




a Shade $A' \cup B$

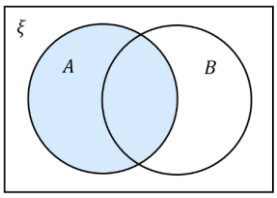


b Shade $A' \cap B$

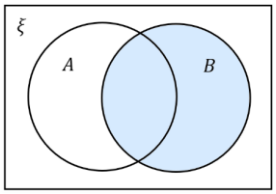


8 Shade the Venn diagrams with the set.

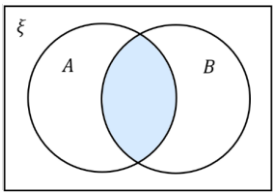
a A



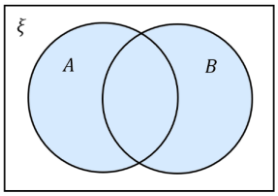
b B



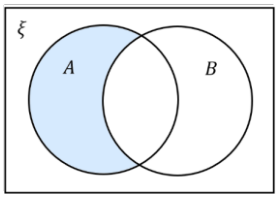
c $A \cap B$



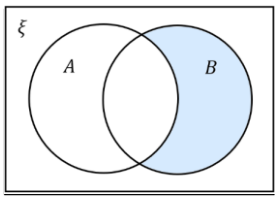
d $A \cup B$



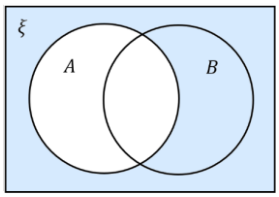
e $A \cap \bar{B}$



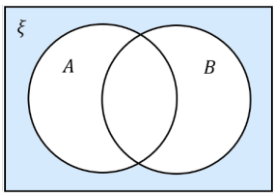
f $\bar{A} \cap B$



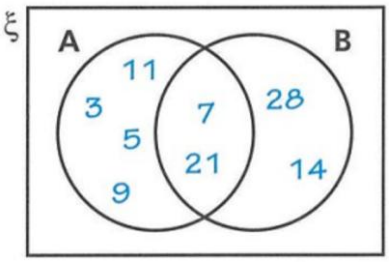
g $\bar{A}$



h $\bar{A} \cap \bar{B}$



9 Here are two sets, A and B.



a Write the elements of $A \cup B$

$A \cup B = \{3, 11, 5, 9, 7, 21, 28, 14\}$

b Write the elements of $A \cap B$

$A \cap B = \{7, 21\}$

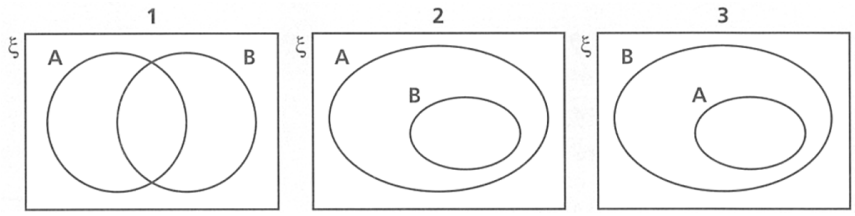
c Write down the elements of $\bar{A} \cap B$.

$\bar{A} \cap B = \{28, 14\}$

d Write down the elements of $A \cap \bar{B}$.

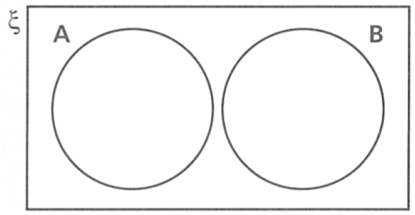
$A \cap \bar{B} = \{11, 3, 5, 9\}$

10 ξ = {1, 2, 3, 4, 5, 6, 7, 8} A = {1, 2, 3, 4, 5} B = {2, 3, 4}
Which of the Venn diagrams would you use to represent these sets?



Venn diagram 2, B is a subset of A

11 Here are two sets represented on a Venn diagram.



a Explain why A ∩ B does not contain any elements.

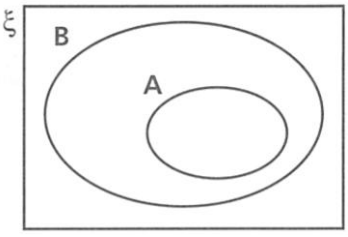
The two sets are disjoint (mutually exclusive)

There are no elements common to A and B

b Give an example of two possible sets A and B.

Even numbers, odd numbers

12 Here are two sets represented on a Venn diagram.



a Explain why A ∩ B is equal to the elements of set A.

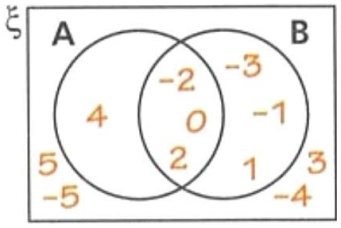
A is a subset of B

So every element of A is in B

b Give an example of two possible sets A and B.

B: even numbers, A: multiples of 4

13 ξ = {numbers from -5 to 5 inclusive}
A = {-2, 0, 2, 4} B = {-3, -2, -1, 0, 1, 2}
Draw a Venn diagram to represent this information.



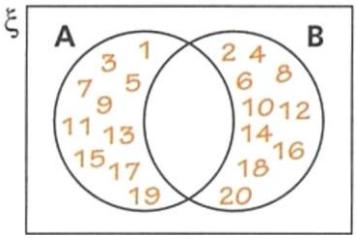
14 ξ = {integers between 1 and 20 inclusive}
List the elements of these sets.

a A = {odd numbers}

b B = {multiples of 2}

c Draw a Venn diagram to represent the information.

d Why are there no elements in both set A and set B?



$A = \{1, 3, 5, 7, 9, 11, 13, 15, 17, 19\}$

B = {2, 4, 6, 8, 10, 12, 14, 16, 18, 20}

No number is both odd and even

15 In a class of 30 students, 18 play soccer, 12 play basketball, and 7 play both sports.
How many students play soccer or basketball?

18 + 12 – 7 = 23

16 In a group of 150 employees, 65 have a car, 85 have a smartphone, and 40 have both.
How many employees have a car or a smartphone?

65 + 85 – 40 = 110

17 A gym has 150 members. 90 use the weights, 110 use cardio equipment.
If all members use at least one type of equipment, how many use both?

$150 = 90 + 110 - x$

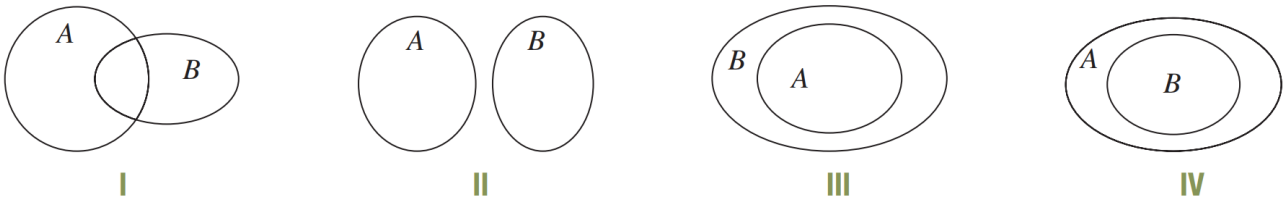
$x = 50$

18 40 children take part in a school show. All the children either sing, dance or do both.
35 children dance. 28 children sing.
How many children sing and dance?

Using addition rule, $40 = 35 + 28 - x$

$x = 23$

19 Select the Venn diagram that best shows the relationship between each pair of sets A and B:



a A = {Tasmania}, B = {states of Australia}

III

b A = {7, 1, 4, 8, 3, 5}, B = {2, 9, 0, 7}

I

c A = {l, e, a, r, n}, B = {s, t, u, d, y}

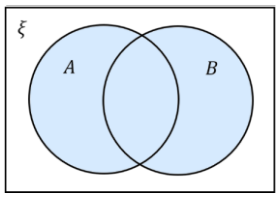
II

d A = {politicians in Australia}, B = {politicians in NSW}.

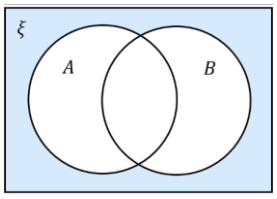
IV

20

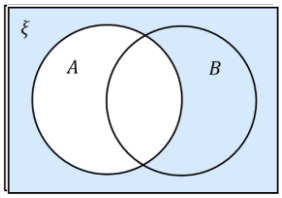
a Shade $A \cup B$



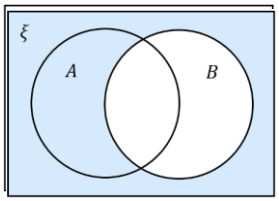
Hence shade $\overline{A \cup B}$



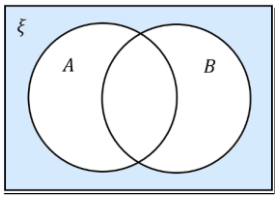
b Shade $\bar{A}$



Shade $\bar{B}$



Shade $\bar{A} \cap \bar{B}$

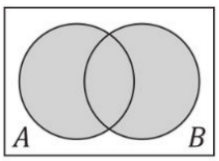


c Hence what can you conclude about $\overline{A \cup B}$ and $\bar{A} \cup \bar{B}$?

They are the same

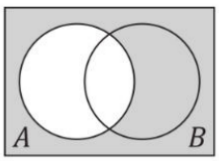
21 Identify the following regions:

a



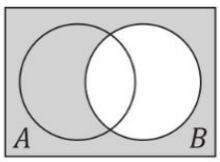
$A \cup B$

b



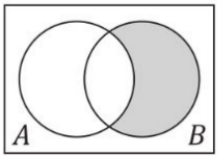
A'

c



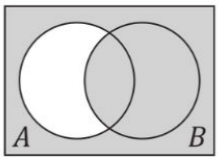
B'

d



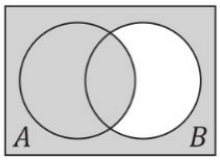
$\bar{A} \cap B$

e



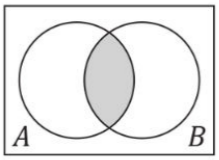
$\bar{A} \cup B$

f



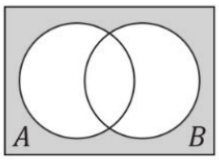
$A \cup \bar{B}$

g



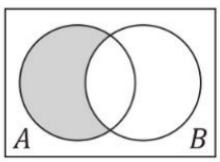
$A \cap B$

h



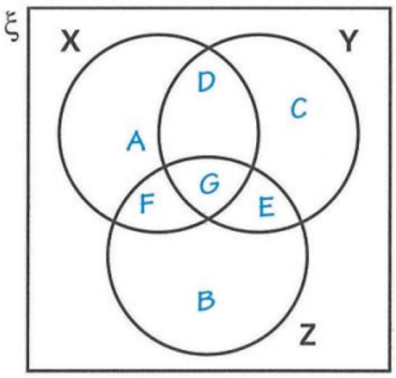
$\overline{A \cup B}$ or $\bar{A} \cup \bar{B}$

i



$A \cap \bar{B}$

22 Here are three sets X, Y and Z. Write down the elements of



a X ∩ Y — {D, G}

b Y ∩ Z — {G, E}

c (X ∩ Y) ∩ Z — G

d Y' — A, F, B

e X' ∪ Y' — {C, E, B, A, F}

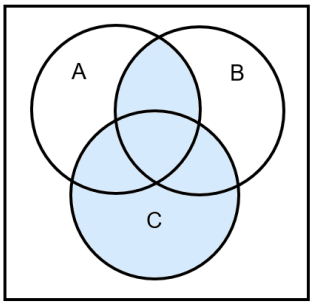
f X ∪ Y' — {A, D, G, F, B}

g (X ∪ Y)' — {B}

h (X ∪ Y) ∩ Z' — {A, D, C}

23

a Shade the region $(A \cap B) \cup C$



b Shade the region $A \cap (B \cup C)$

